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“The sole objective of the investigation of an accident under the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012 shall be the prevention of future accidents through the ascertainment of its causes and circumstances. It shall not be the purpose of an such investigation to determine liability nor, except so far as is necessary to achieve its objective, to apportion blame.”

NOTE

This report is not written with litigation in mind and, pursuant to Regulation 14(14) of the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012, shall be inadmissible in any judicial proceedings whose purpose, or one of whose purposes is to attribute or apportion liability or blame.

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Foundering of fishing vessel

Kairos (BF190)

70 nautical miles west of the Isles of Scilly

18 May 2015

SUMMARY

At about 2335¹ on 18 May 2015, the 18.6m twin-rig prawn trawler *Kairos* capsized and foundered in very rough seas 70 nautical miles west of the Isles of Scilly, UK. The vessel's five crew abandoned into the water and climbed into a liferaft. They were rescued by a nearby fishing vessel. *Kairos*'s skipper fell into the sea when leaving the liferaft but he was quickly recovered from the water. There were no injuries and there was no pollution.

The capsize occurred after *Kairos* had developed a significant port list while attempting to recover a lost trawl. It is likely that the vessel's total loss of stability was primarily due to:

- The repeated swamping of the aft deck by high waves.
- The load on the creeper wire used to try and recover the lost net.



Image courtesy of Sean Boyce and Marine Traffic

Kairos

¹ All times in this report are Co-ordinated Universal Time (UTC) +1 unless stated otherwise

- The downflooding of sea water through the cabin space air vent (after the port list reached an angle of about 35°).
- The downflooding of sea water through an engine room air vent (after the port list reached an angle of about 48°).

The dangers of trawling and recovering lost gear are promulgated in Marine Guidance Note 415 (F). The MAIB has published a flyer to the fishing industry highlighting this guidance and the vulnerability of fishing vessels operating in extreme or unusual circumstances. Consequently, no recommendations have been made.

FACTUAL INFORMATION

Narrative

On 18 May 2015, *Kairos* was trawling 70nm to the west of the Isles of Scilly (**Figure 1**) with a relief skipper in command. At approximately 1630, the starboard net's pennant ropes parted as the net was being hauled. The net, which weighed 1.4t, was lost and *Kairos's* skipper marked the vessel's position on the electronic chart plotter.

The port trawl was hauled and stowed. A creeper² was then rigged on a wire leading from the centre block (**Figure 2**) and lowered to the seabed. *Kairos* was then manoeuvred ahead in the vicinity of the position marked on the plotter to try and snag the lost net with the creeper. The vessel's two deckhands and two engineers tidied the net drum space. They then went inside the accommodation on the main deck. *Kairos* was rolling heavily in 4m seas and 30kts west-north-westerly winds. At 1717, the skipper advised the vessel's regular skipper by text message³ that he was attempting to find a lost net.

At some time between 2200 and 2230, the tone of *Kairos's* engine changed and the vessel slowed. The winch holding the creeper wire also rendered. The creeper had snagged on an obstruction, possibly the lost net, so the skipper immediately reduced the engine speed to between 500 and 600rpm⁴. The vessel was on an easterly heading. By now, the height of the swell waves had increased to over 5m and the wind was gusting up to 40kts.

The change in the engine's tone also alerted *Kairos's* crew, who donned foul weather clothing and personal flotation devices and then went to the aft weathertight door (**Figure 3**) leading to the net drum space. As a deckhand opened the door, a wave broke over the stern and water flooded through the door opening into the alleyway of the main deck accommodation. The door was closed and secured with six dog clips. The crew informed the skipper of the water ingress and then started bailing the floodwater with buckets through the forward scuppers.

At about 2235, *Kairos's* skipper telephoned the vessel's regular skipper via a satellite link. He informed him that he had found the net but that there was a small amount of water in the accommodation. He also informed him that there was no cause for concern. At 2238, the regular skipper sent *Kairos's* relief skipper a text message advising that the engine should be run at 1000rpm, otherwise the quarter would

² A creeper is a large grappling hook type device. *Kairos's* creeper was approximately 1.5m long and weighed about 80kg. The wire through the centre block was 22mm diameter steel wire rope that had a breaking load of approximately 30t.

³ *Kairos* was equipped with a satellite data system that allowed the skipper to access the internet and communicate by smartphone.

⁴ Revolutions per minute. *Kairos's* maximum engine speed was about 1800rpm.

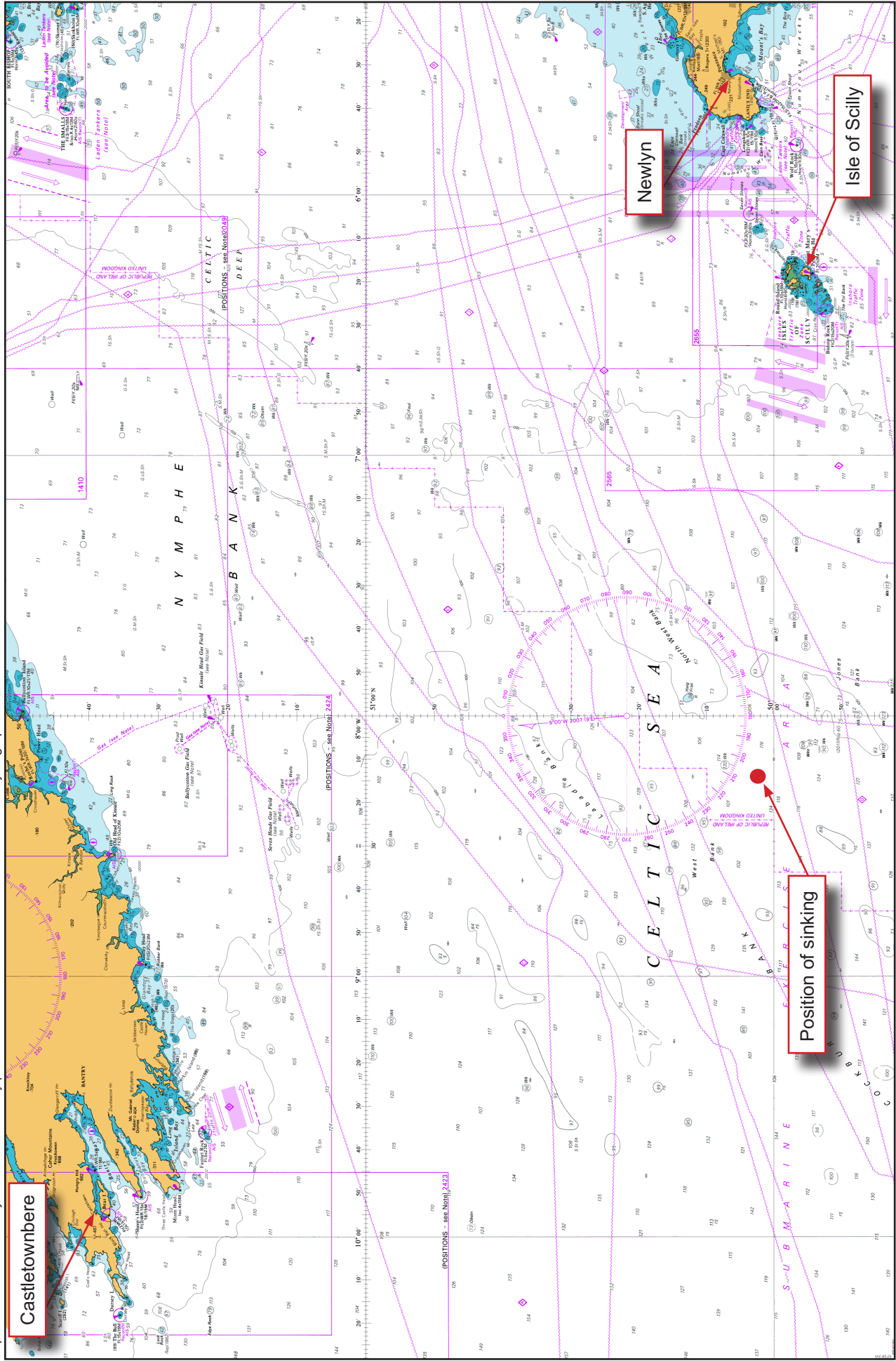


Figure 1: Extract from Admiralty chart 1123



Figure 2: Stern of Kairos (as Vision II)

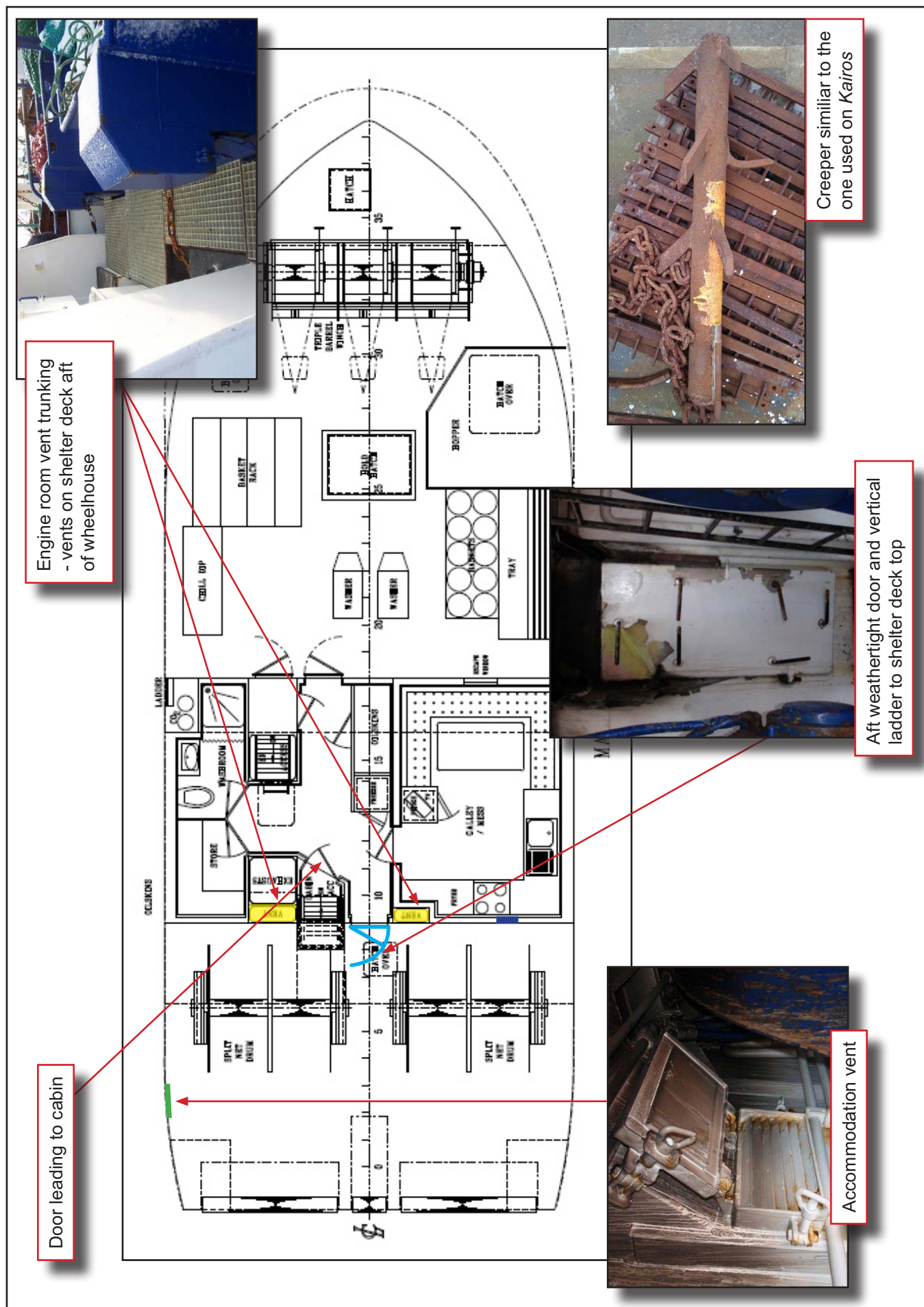


Figure 3: Vents and aft door positions

become “watery”. The message was received on the skipper’s smartphone but it was not read⁵. The engine speed remained at between 500 and 600rpm and *Kairos* maintained an easterly heading with waves continuing to break over the stern into the net drum space.

One of *Kairos*’s engineers joined the skipper in the wheelhouse to operate the creeper wire winch. The creeper wire was slowly heaved. However, the wire soon tightened so it was immediately veered to prevent the creeper from damaging the net. To change the direction of pull on the creeper wire, *Kairos*’s skipper repeatedly applied full starboard rudder and increased engine power as the creeper wire was veered. The skipper was attempting to turn the vessel through south and into wind, but the vessel’s heading would not turn beyond 170° (**Figure 4 positions 1-4**). The heading soon returned to an easterly direction when the engine speed was reduced and the creeper wire was heaved (**Figure 4 positions 5-8**). The vessel’s port list was now about 35° and increasing.

Meanwhile, a crewman on the main deck opened the door at the top of the stairs leading to the cabin space below (**Figures 3, 5 and 6**) and saw approximately 1m of water at the bottom of the stairwell. The skipper was informed of the flood and he ordered the crew to pump the water out with the portable electric-driven pump. The engine room and fish holds were dry, but as a precaution the compartments’ bilge pumps were started.

Kairos’s crew became increasingly concerned as the port list continued to increase and the vessel was still rolling heavily. Also, although the electric-driven pump was pumping water from the cabin space, the water level was still rising and the skipper’s attempts to turn the vessel continued to be unsuccessful. The crew repeatedly told the skipper to release the creeper wire.

By about 2315, the skipper acknowledged that the situation was serious. He instructed the crew to put on their abandon ship lifejackets. Via VHF radio, he also informed the skipper of *Vision III*⁶, which was 20nm away, that ‘he was having a few problems and that the crew were screaming at him to let the wire go’. *Kairos*’s skipper requested that *Vision III* close in case he required assistance. It is reported that the skipper also took action to release the creeper wire.

At about the same time, *Kairos*’s skipper saw on the closed circuit television monitor that the atmosphere in the engine room was ‘misty’. He continued to try and turn the vessel to the south and told the crew to prepare the liferafts⁷. He also informed *Vision III*’s skipper via medium frequency (MF) radio that *Kairos* was probably going to capsize, and requested that he broadcast a “Mayday” relay⁸. *Vision III*’s skipper transmitted radio “Mayday” calls by voice on MF 2182kHz and VHF channel 16.

Kairos’s skipper realised that the vessel was close to capsize. He left the wheelhouse and told the crew to launch the liferafts. The crew had already released the liferafts’ lashings, but they were unable to lift the liferafts from their cradles and launch them overboard because they could not keep their footing on the listing deck. Moments later, at about 2335, the engine stopped and the vessel started to capsize.

The skipper and the crew abandoned into the water. As they did so, one of the crewmen pulled and then held onto one of the liferafts’ painters. The liferaft inflated and *Kairos*’s crew climbed on board. Shortly

⁵ The phones used were smartphones, with a text application that enabled a person sending a text message to see when the message had been received and read by the recipient.

⁶ *Vision III* and *Kairos* shared ownership and management interests through Westward Fishing Company Ltd.

⁷ *Kairos* carried two liferafts aft of the wheelhouse on the centreline.

⁸ A “Mayday” relay message can be sent by any vessel if there is grave and imminent danger to a person, vessel or aircraft and immediate assistance is required.

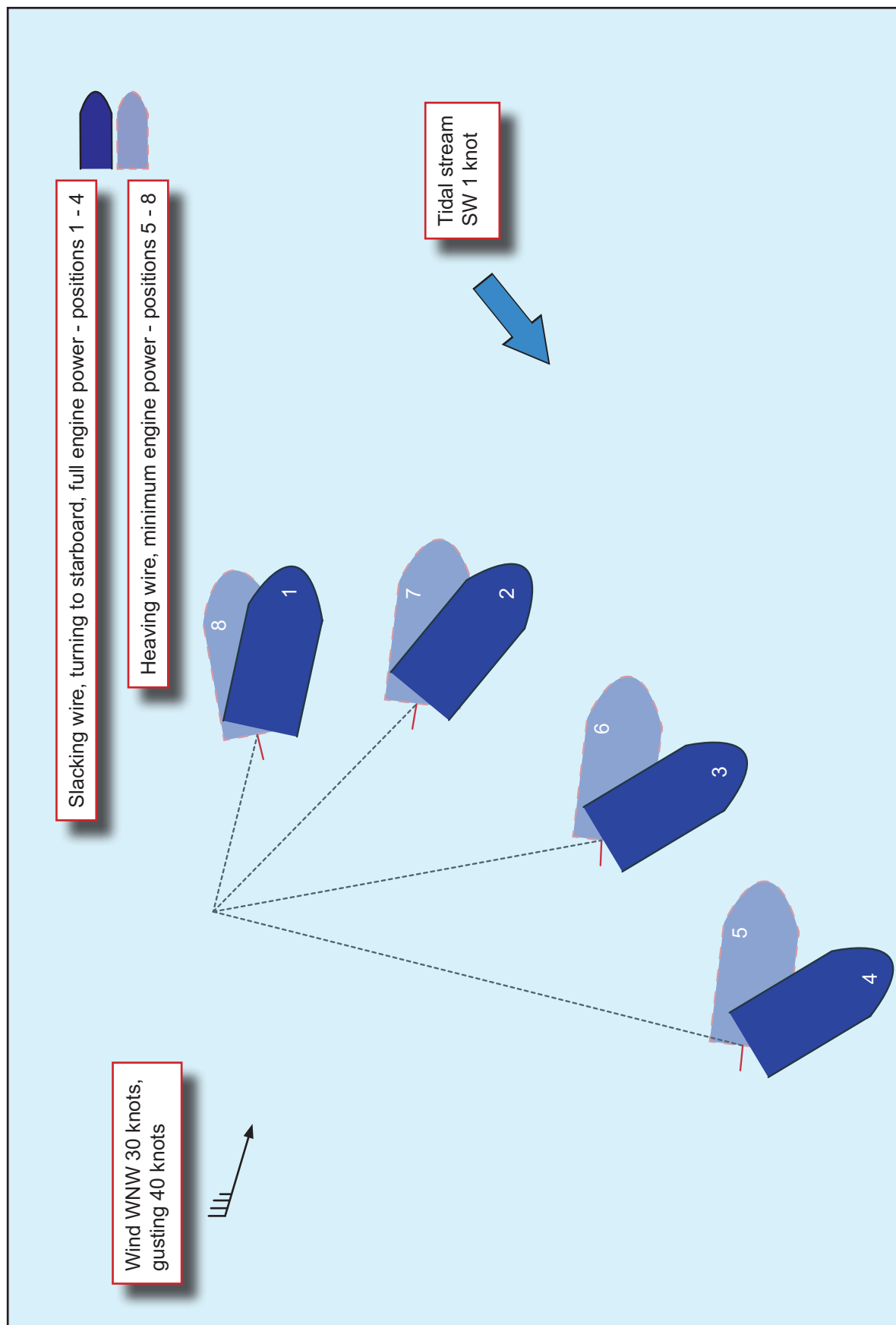


Figure 4: Diagram showing movement of *Kairos* while veering and heaving centre trawl wire

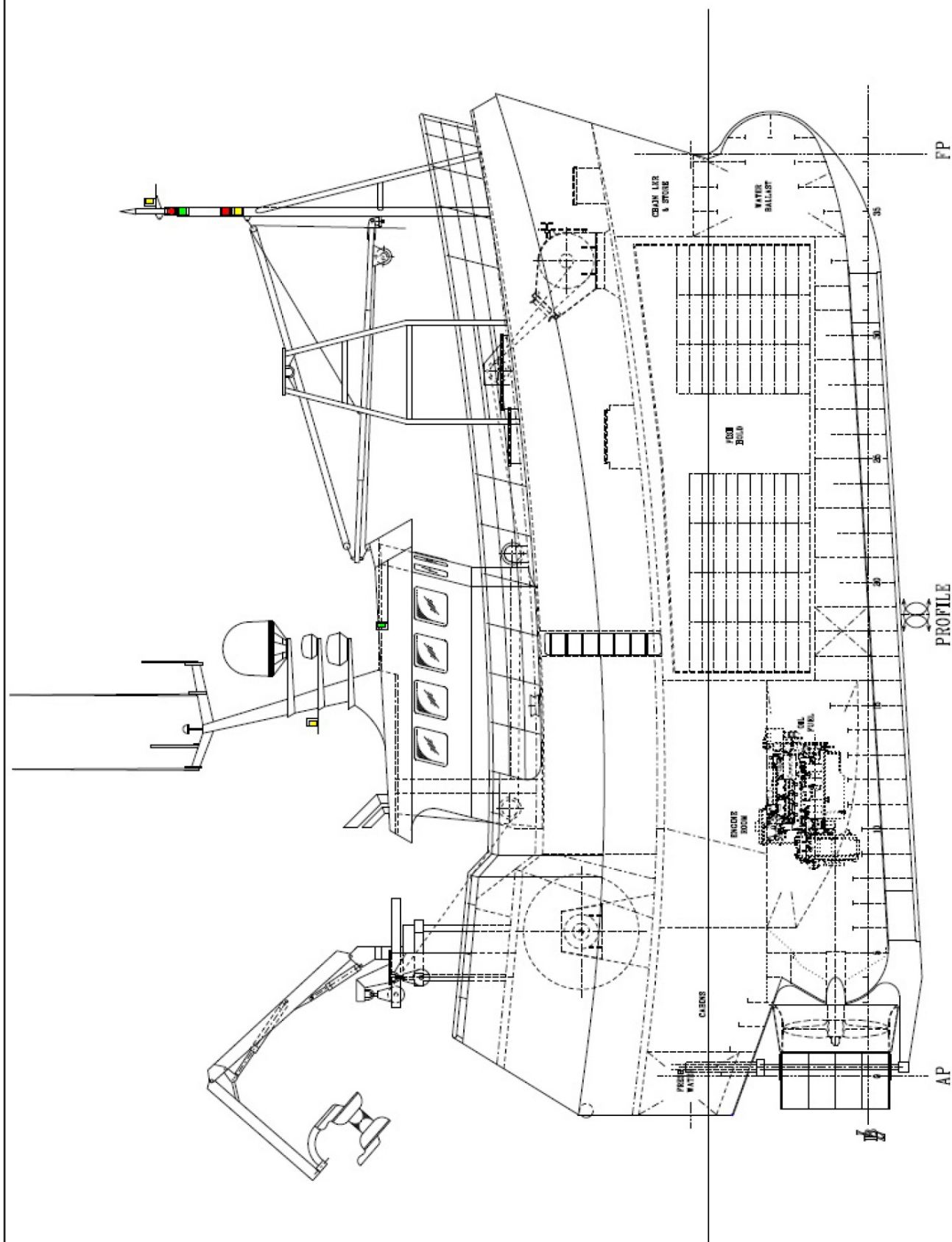
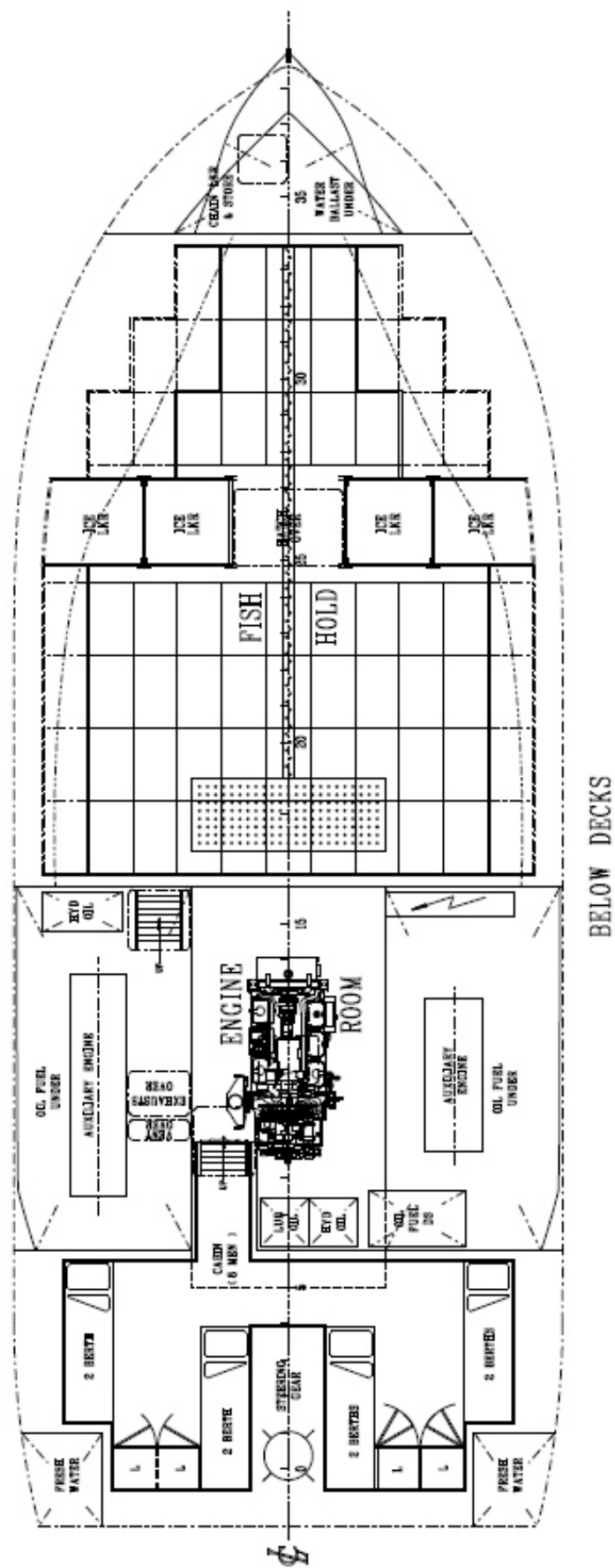


Figure 5: Side elevation of *Kairos*



BELOW DECKS

Figure 6: Plan below decks

afterwards, *Kairos* sank in a depth of about 100m. The vessel's EPIRB⁹ surfaced and transmitted a distress alert. The second liferaft also inflated. At about 2346, *Vision III*'s skipper broadcast distress messages via MF and VHF DSC¹⁰.

Recovery

The EPIRB and MF distress alerts were received by Falmouth Coastguard, which contacted *Kairos*'s owner and alerted vessels in the area. The Irish registered fishing vessel *Cu Na Mara* was 5nm north of the EPIRB and closed to assist. At 0100, *Cu Na Mara* was manoeuvred alongside the manned liferaft and *Kairos*'s crew boarded the fishing vessel by ladder.

Kairos's skipper was the last to leave the liferaft. He was cold and wet and was holding on to the ladder as it was pulled on board. However, the skipper's arm became squashed against the deck guardrails, causing him to let go of the ladder and fall into the water. The skipper was thrown a lifebuoy but he was unable to hold onto it. One of *Cu Na Mara*'s crew donned an immersion suit and a safety line. Supervised by his skipper, the crewman jumped into the water. He then secured a rope from a derrick around *Kairos*'s skipper, who was then hoisted on board. *Cu Na Mara*'s crewman was also recovered from the water and the vessel made best speed towards Castletownbere, Ireland, arriving later the same day.

Construction

Kairos was a steel-hulled twin-rig trawler that was built in 2004 and met the requirements of The Code of Safe Working Practice for the Construction and Use of 15 metre length overall to less than 24m registered length Fishing Vessels (MSN 1770 (F)). The vessel was originally named *Amethyst* but, following several name changes, was re-named from *Vision II* to *Kairos* in March 2015.

Three main compartments were located below the main deck: the fish hold, the engine room and a cabin space (**Figure 6**). The fish hold's forward bulkhead, the engine room's aft bulkhead and the common bulkhead between the fish hold and the engine room were watertight. A steering trunk containing the rudder stock was located on the centreline at the aft end of the cabin space. The steering trunk's internal bulkheads were not watertight.

Pumping arrangements

Kairos was fitted with two fixed bilge pumps in the engine room, each with 25m³/hr capacity, which could also be used to pump out the fish hold through a valve chest. The vessel also carried a petrol-driven salvage pump (25m³/hr) and an electric-driven emergency submersible pump (12m³/hr). Both the engine room and the fish hold were fitted with bilge alarms.

The cabin space was fitted with a spring-loaded manual valve that enabled small amounts of water to drain to the engine room. The chain locker forward of the fish hold was fitted with a manual pump. Neither the cabin space nor the chain locker was fitted with water level alarms.

Air vents

Two engine room air vents (0.13m x 0.51m) were sited 0.61m above the deck, aft of the wheelhouse (**Figure 3**). A further engine room air vent (0.6m x 0.27m) was sited 1.57m above the deck on the aft end of the funnel. The cabin space air vent (0.13m x 0.42m) was sited 1.96m above the trawl deck on the port side of the net drum space (**Figure 3**). At the time of the accident all of the air vents were open.

⁹ Emergency Position Indicating Radio Beacon

¹⁰ Digital Selective Calling or DSC is a standard for sending pre-defined digital messages via the medium frequency (MF), high frequency (HF) and very high frequency (VHF) maritime radio systems.

Stability booklet

The working instructions detailed in *Kairos's* stability booklet, which was approved on 21 January 2011, stated that openings leading to watertight spaces should be closed when not in use. The instructions also stated:

*In certain circumstances and/or conditions such forces may be exerted by the combined effects of the winches and propulsion machinery as to bring about CAPSIZE of the vessel. Care should be taken when hauling gear and shipping the catch, unequal loading on warps may generate significant list angles. Particular care should be taken when freeing snags, or when towing warps are close to the vertical and the winches are operating at significant load. **The stability of the vessel when operating in these conditions may be substantially reduced.*** [sic]

The stability booklet contained details of *Kairos's* freeing ports, vents and openings. The lower engine room vents (**Figure 4**) were identified as downflooding points in all loading conditions (as the engine could not run with them closed) at angles between 57.5° and 66.6°.

Crew

The skipper was 46 years of age and was a United Kingdom national. He had been fishing all his working life and held a fishing vessel Class 2 certificate of competency. The skipper had worked as mate on other similar vessels and had worked as relief skipper on one of these for 1½ years.

Kairos's skipper joined the vessel as a relief skipper in Newlyn, UK during the afternoon of 13 May 2015. It was his first time on board the vessel and he familiarised himself with its fishing gear and method during the time he spent on board prior to the vessel's departure the following day. The skipper expected the trip to last for up to 6 days. For the first 3 days, *Kairos* operated out to 40nm west of the Isles of Scilly. However, the catches were poor so the skipper moved the vessel further offshore.

The four crew were Filipino nationals who were career fishermen and had worked on fishing vessels for many years. Two of the crewmen were STCW¹¹ qualified engineering watch ratings and two were STCW qualified navigation watch ratings. All of the crew had attended the mandatory training courses or recognised equivalents required by fishermen working on board UK registered vessels¹². The crew joined *Kairos* in November 2014 for a 10-month contract; three had worked on board previously.

Hindcast

A marine weather data hindcast report covering the period and location of the accident was provided by the Met Office. The report included:

With a significant wave height of 5.3 metres, consisting of predominately wind wave from a west-northwesterly direction, the seas could be described as being very rough. With a zero upcrossing period¹³ of 8 seconds, a maximum wave height of 1.7 times significant wave height or 9.0 metres could have been engaged during a 60 minute interval between 2200 and 2300 UTC on the 18th May 2015. [sic]

¹¹ Seafarers Training Certification and Watchkeeping certificate for skippers of fishing vessels in limited areas.

¹² New entry fishermen must complete basic safety courses in sea survival, elementary first-aid; fire-fighting and health and safety. Fishermen with 2 years of experience must also complete a 1 day safety awareness course run by the Sea Fish Industry Authority (Seafish)

¹³ Wave period

ANALYSIS

Stability assessment

A computer-based stability assessment was undertaken to determine the causes and circumstances of *Kairos*'s reduction of stability, which initially manifested as a port list and eventually led to the vessel's capsize and loss. The assessment used data from the vessel's stability booklet and, based on the evidence available, it assumed that the fuel and water tanks were at 50% of capacity, all weather tight doors were closed and all vents and freeing ports were open and free from obstruction. It is not known whether the cross connection valve between the port and starboard fuel tanks was open or closed.

With regard to *Kairos*'s loaded condition prior to the attempted recovery of the lost net, the stability assessment identified that:

- The vessel met all of the applicable stability criteria provided that the initial downflooding points were the engine room vents (**Figure 2**) (with the vessel at an angle of 60° list) and that the fuel tank cross-connection valve between port and starboard tanks was closed.
- The vessel did not meet six out of seven stability criteria with the cabin space vent open (**Figure 2**). The open vent reduced the angle of downflooding to 35°.

A number of vessel conditions were then assessed to reflect the circumstances that led to *Kairos*'s capsize. These conditions included: 5t of floodwater in the net drum space (**Figure 2**), floodwater depths of 0.61m, 1.53m and 2m in the cabin space and loads of 2t, 4t and 6t on the trawl wire. The resulting assessments identified:

- *Kairos* retained positive stability with the varying levels of water in the cabin space and 5t of water in the net drum space.
- Wire loads of 2t, 4t and 6t would have listed the vessel to 22°, 34° and 44° respectively, the latter bringing the vessel very close to capsize.
- With 2m depth of water in the aft cabin space and 5 tonnes of water in the net drum space, the angle of downflooding through the port engine room vent into the engine room would still not have occurred until 48° of heel/list.
- With the fuel tank cross-connection valve open the free surface moment of the combined fuel tank contents increased from 6tm¹⁴ to nearly 72tm.

Flooding

The floodwater in the cabin space could only have entered from a hull penetration below the waterline within the space or the steering trunk, the compartment's air vent, or the adjacent engine room. As the engine room was checked and found to be dry shortly after the flood in the cabin space was detected, and the bulkhead between the engine room and the cabin space was watertight, this source can be eliminated. The water ingress into the cabin space was due either to downflooding through the vent or a hull penetration.

With the creeper fast on the seabed, it is feasible that the creeper wire could have snagged and damaged the rudder stock and its associated through-hull fitting, resulting in water ingress. However, the wind and waves and the ahead direction of the main engine would have continually moved *Kairos* away

¹⁴ Tonnes metre

from the creeper wire. In such circumstances, it would have been extremely difficult for the scope of the wire to be shortened sufficiently to bring it into contact with the rudder by the operation of the winch alone.

It is more likely that the water ingress into the cabin space was from downflooding through the air vent sited on the port side of the net drum space (**Figure 3**). *Kairos* was anchored by the creeper and its wire and was stern-to the very rough, short, high seas. Consequently, waves potentially as high as 9m would have repeatedly 'pooped' the fishing vessel as it pitched, rolled and heaved. It is unlikely that the freeing ports in the net drum space would have been able to clear such a volume of water. Therefore, the net drum space would have been swamped up to the height of the bulwark with the weight of the water trimming the vessel by its stern. In such circumstances, it is likely that the cabin space vent would have been immersed for prolonged periods even before the 35° immersion angle was reached.

Nonetheless, the flooding of the cabin space alone would not have sufficiently reduced the vessel's stability to cause it to capsize or founder. With the space completely flooded, *Kairos* would have still retained positive stability and would have been able to proceed to safety.

Listing and capsize

When floodwater was discovered in the cabin space, *Kairos's* skipper had been attempting to recover the lost starboard net for over 1 hour. During this time, the vessel had developed a significant port list. The evidence available, along with the findings of the subsequent stability assessments, indicates that this list was most likely caused by the combined load put on the creeper wire by the winch, the sea conditions and the skipper's attempts to turn the vessel. A load of just over 4 tonnes on the creeper wire was sufficient to induce the 35° list required for downflooding through the cabin space vent in static conditions. It was reported that the valve on the pipe connecting the port and starboard fuel tanks was normally kept closed. However, if it had been left open the angle of list could also possibly have been exacerbated by the free surface effect of fuel running between the tanks.

Over time, the situation would have deteriorated. Although the creeper wire was slackened, it would have soon re-tensioned in the prevailing conditions. With the continuing, and possibly increasing, load on the creeper wire, the deepening floodwater in the cabin space would have increased the angles of trim and list due to its weight and its free surface effect as *Kairos* rolled. In turn, the increasing angles of trim and list would have led to the deeper immersion of the cabin space vent and a faster rate of downflooding. By the time the crew attempted to pump the floodwater from the cabin space, the water ingress was evidently greater than the capacity of the electric-driven pump.

As soon as the angle of list had increased beyond 48°, the immersion angle of the engine room air vents, downflooding through the port engine room vent would have occurred. This is supported by the presence of mist in the engine room and also possibly by the activation of bilge alarms in the minutes before the crew abandoned. It was reported that the skipper took action to release the creeper wire. However, the evidence available indicates this was not achieved and the wire remained connected throughout until, ultimately, the flooding of the engine room resulted in *Kairos's* capsize.

Decision-making

The hazards associated with recovering lost gear were highlighted in *Kairos's* stability booklet. The Maritime and Coastguard Agency has also promulgated these hazards to the fishing industry in Marine Guidance Note 415 (F), which states:

5.2 Recovery of fouled gear can impose extra loads on wires and machinery, particularly in adverse weather conditions. Failure of either may result in excessive rolling or a dangerous list to the vessel.

5.3 The vessel's stability reserves may be seriously reduced when hauling on fouled gear with the winches working hard. Additionally winches should not be braked and used in conjunction with a vessel's motions to free fouled gear; a heavier than normal swell may be sufficient to bring about the vessel's capsize in this condition.

Kairos capsized while the skipper attempted to recover the starboard net with the creeper in very rough seas. The skipper's perseverance in this respect, in worsening sea conditions, deepening floodwater in the cabin space, a significant and increasing port list and his crew shouting at him to release the creeper wire, indicate that he did not properly consider the hazards detailed in MGN 415(F) and that he was extremely slow to recognise the perils of his actions. This allowed *Kairos's* stability to diminish until its loss was inevitable. *Kairos* was a stable vessel, and running the creeper wire off the winch would have immediately reduced the port list and prevented capsize.

Kairos's relief skipper was new to the vessel and he was not familiar with its operation. He was also unaware of the potential for downflooding through the cabin space vent. Moreover, his priorities were likely to have been influenced by the commercial implications of losing the net and, had it been released, the trawl wire and creeper, on top of the vessel's already poor catches.

Recovery of the skipper

There have been numerous accidents in which fishermen have fallen overboard and have died as a result of crews being unable to recover them from the water.¹⁵ Although *Kairos's* skipper was wearing a lifejacket when he fell from the ladder while boarding *Cu Na Mara*, he was cold and tired and therefore in danger. The quick-thinking and prompt action taken by *Cu Na Mara's* crew to safely recover him from the very rough seas in the dark was commendable.

CONCLUSIONS

- *Kairos's* capsize resulted directly from the skipper's continued attempts to recover the starboard net in very rough seas.
- The vessel was anchored to the seabed by the creeper wire and the net drum space was repeatedly swamped by heavy seas.
- The load on the creeper wire induced a significant and increasing port list.
- It is probable that the flooding in the cabin space was caused by downflooding through an air vent.
- The prompt action taken by *Cu Na Mara's* crew to recover *Kairos's* skipper from the water was commendable.

¹⁵ See MAIB report 26/2015 Report on the loss of a person overboard from the fishing vessel Beryl (BF 440) with the loss of one life west of the Shetland Islands on 10 February 2015

ACTION TAKEN

MAIB actions

The MAIB has issued a flyer to the fishing industry highlighting the advice provided in MGN 415 (F) and the vulnerability of fishing vessels operating in extreme or unusual circumstances.

RECOMMENDATIONS

In view of the flyer and other guidance currently promulgated to the fishing industry, no recommendations have been made.

SHIP PARTICULARS

Vessel's name	<i>Kairos</i>
Flag	United Kingdom
Fishing number	BF 190
Type	Twin rig prawn trawler
Registered owner	Jack Fishing LLP
Manager/agent	Westward Fishing Company
Year of build	2004
Construction	Steel
Length overall	18.6m
Registered length	17.7m
Gross tonnage	163t
Main engine	Diesel - Caterpillar 3412
Power	217kW

VOYAGE PARTICULARS

Port of departure	Newlyn, Cornwall, UK
Type of voyage	Coastal
Cargo information	Not applicable
Manning	5

MARINE CASUALTY INFORMATION

Date and time	18 May 2015, 2335 (UTC+1)
Type of marine casualty or incident	Very serious marine casualty
Location of incident	50°02'N 008°11'W
Injuries/fatalities	None
Damage/environmental impact	Vessel lost
Ship operation	Fishing
Voyage segment	Mid-water
External & internal environment	Wind: WNW 30kts - 40kts. Very rough sea. The tidal stream was setting to the south-west at a rate of 1kt. It was dark and the visibility was good.
Persons on board	5